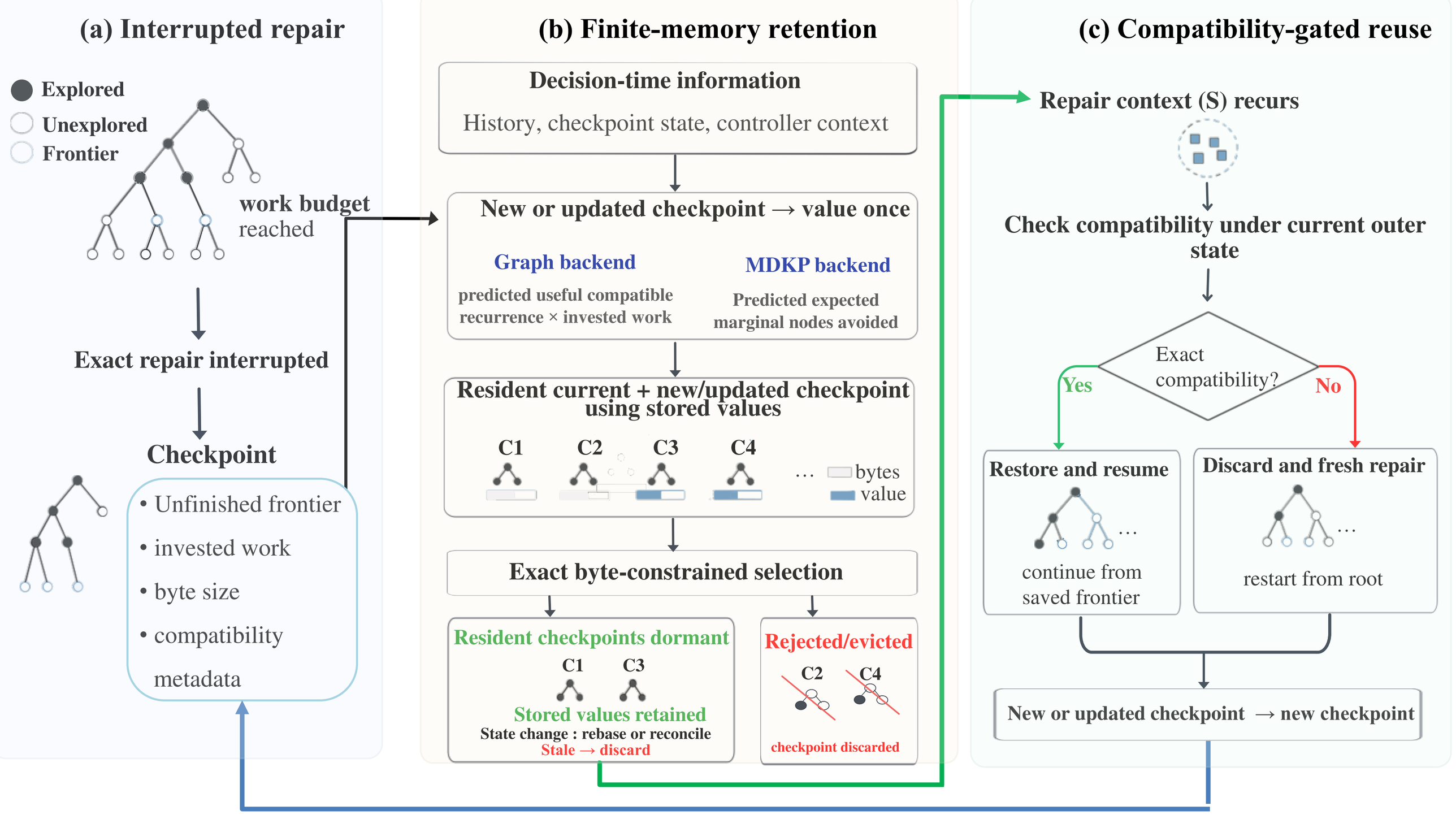

(a) Interrupted repair
Explored
Unexplored
Frontier
work budget reached
Exact repair interrupted
Checkpoint
• Unfinished frontier
• invested work
• byte size
• compatibility metadata
(b) Finite-memory retention
Decision-time information
History, checkpoint state, controller context
New or updated checkpoint → value once
Graph backend
predicted useful compatible recurrence × invested work
MDKP backend
Predicted expected marginal nodes avoided
Resident current + new/updated checkpoint using stored values
C1
C2
C3
C4
bytes
value
Exact byte-constrained selection
Resident checkpoints dormant
C1
C3
Stored values retained
State change : rebase or reconcile
Stale → discard
Rejected/evicted
C2
C4
checkpoint discarded
(c) Compatibility-gated reuse
Repair context (S) recurs
Check compatibility under current outer state
Exact compatibility?
Yes
No
Restore and resume
continue from saved frontier
Discard and fresh repair
restart from root
New or updated checkpoint → new checkpoint